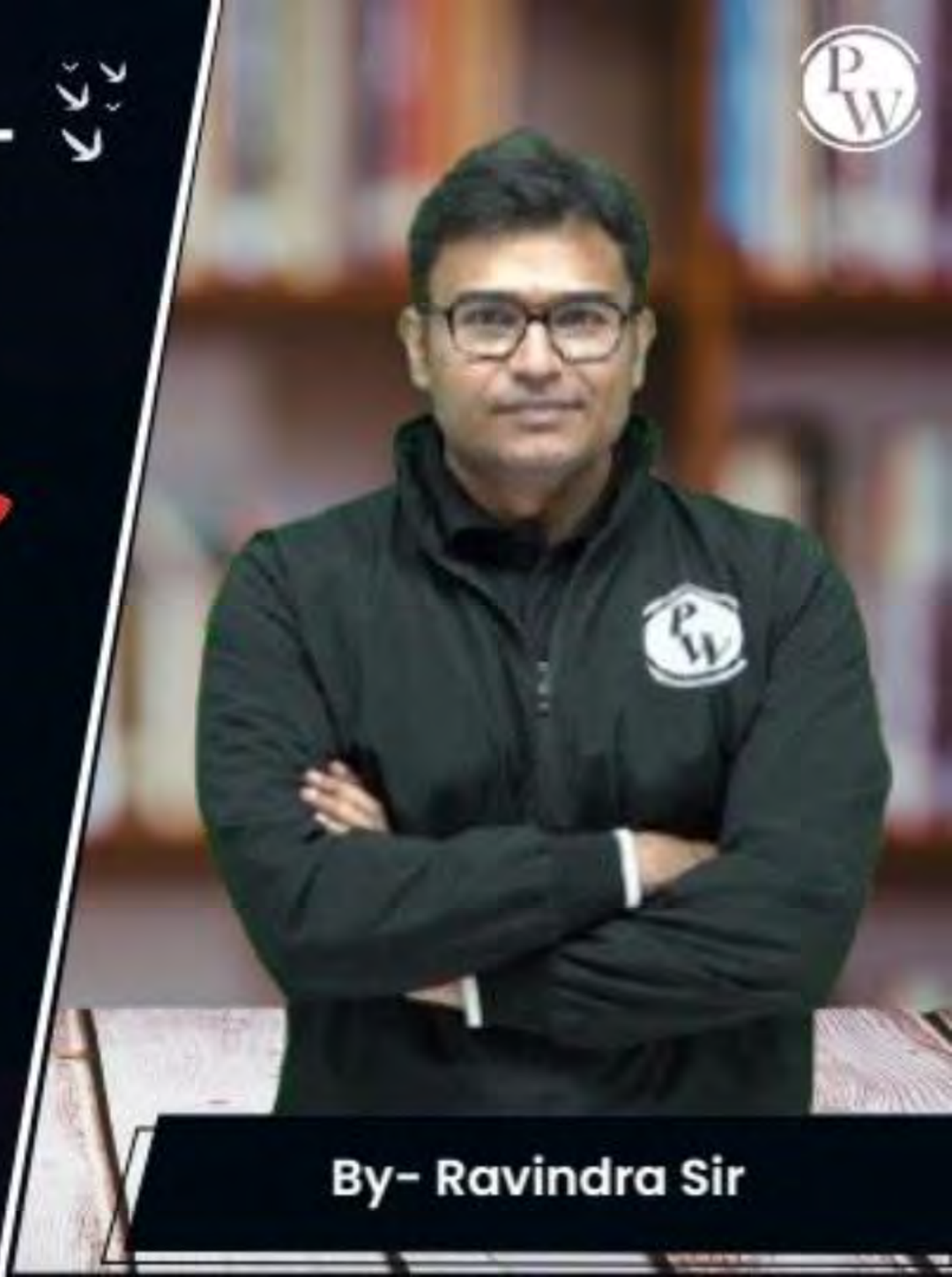
Computer Science & IT

Theory of Computation



Regular Languages

Lecture No. 11



By- Ravindra Sir

Recap of Previous Lecture



Topic — Regular grammars

Topic — CFG

Topic

Topics to be Covered

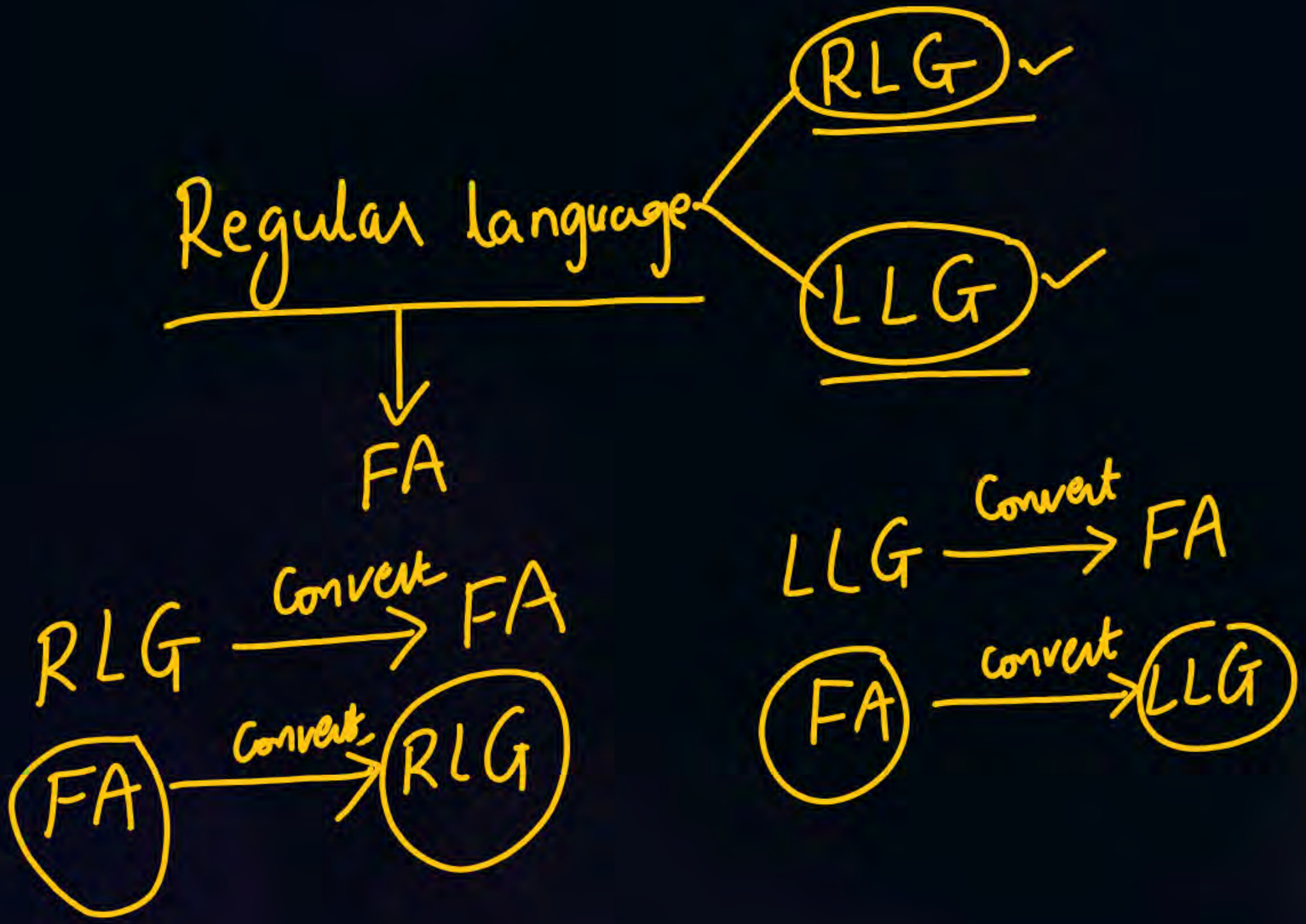


Topic — Regular grammars

Topic — NFA → DFA

Topic — minimisation of DFA

ϵ NFA → NFA


RLG
Regular language
LLG
FA
RLG
Convert
FA
FA
Convert
RLG
LLG
Convert
FA
FA
convert
LLG


RLG
LLG

$S \rightarrow 01S/110$



$X(01)^*(110)$

any no of 01's followed by 110

$(01+1)^*110$

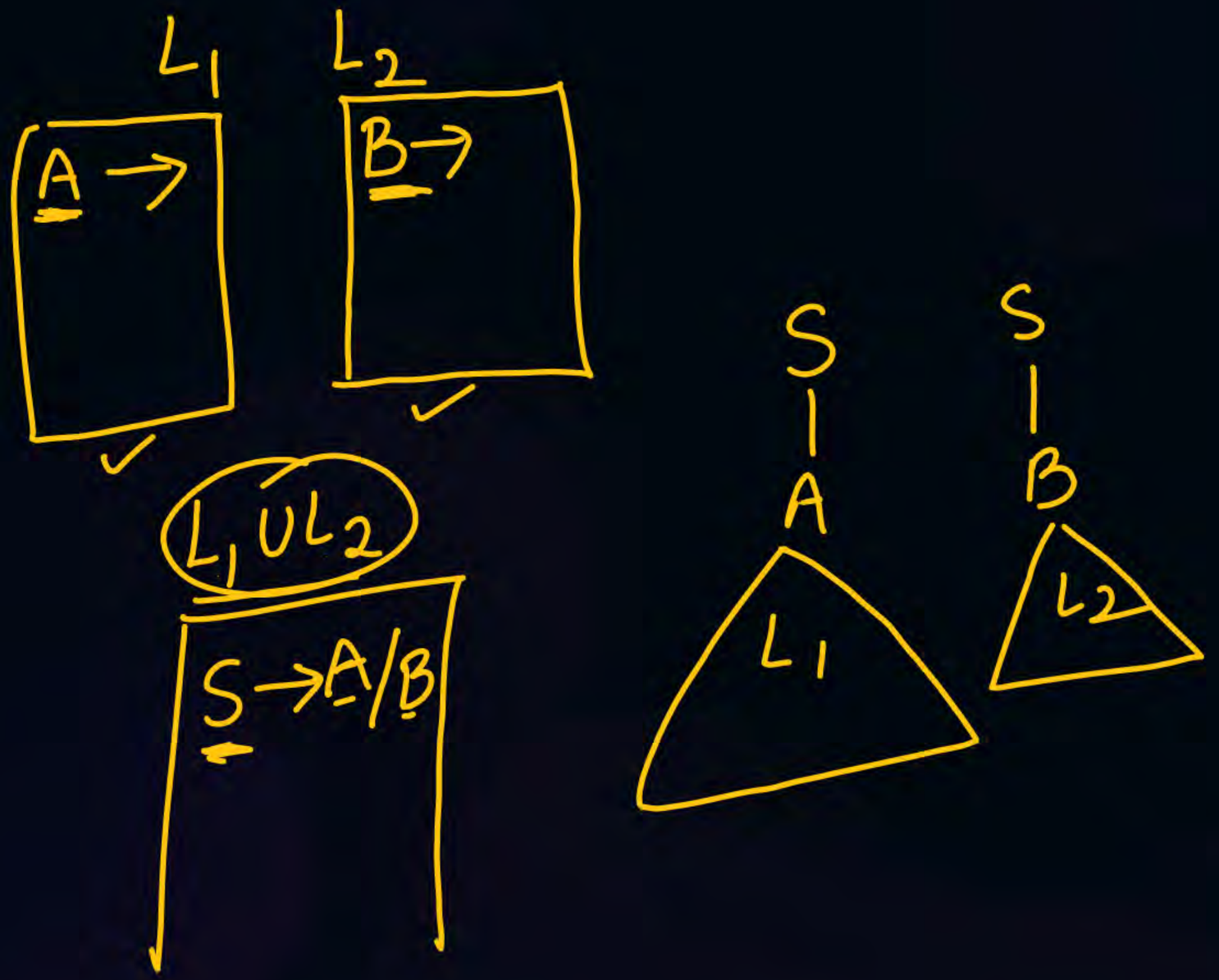
$(a+b)^* = (a^*b^*)^* = (b^*a^*)^*$

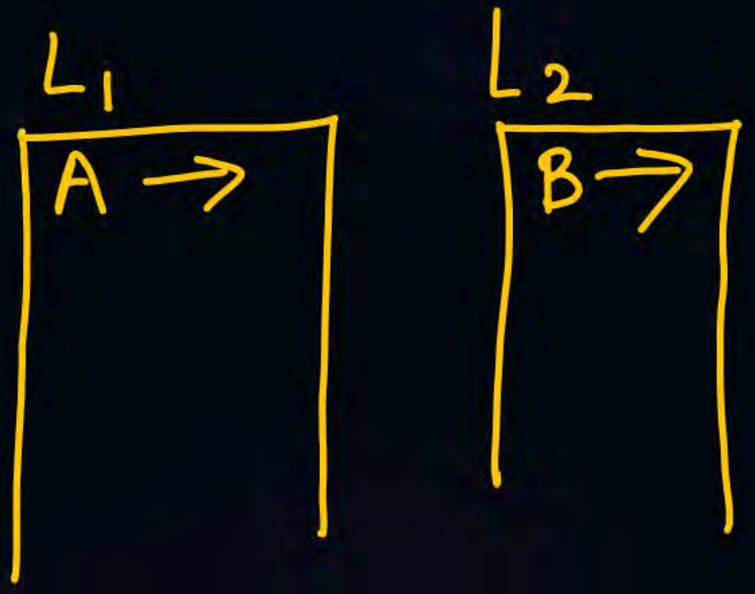
$(a^*+b)^* = (a+b^*)^*$

$(a^*b^*)^*$

$a^*b^*a^*b^*$ $a^*b^*a^*b^*$

a b a b


L1
L2
A →
B →
L1 UL2
S → A/B
S
I
A
L1
S
I
B
L2



$L_1 \cdot L_2$

$S \rightarrow AB$

$a^* + b^* \neq (a+b)^*$

$L_1 \quad L_2$

$S \to S_1 / S_2$

$S_1 \to aS_1 / \epsilon \Rightarrow a^*$

$S_2 \to bS_2 / \epsilon \Rightarrow b^*$

} RLG

$S \to S_1 / S_2$

$S_1 \to S_1 a / \epsilon \Rightarrow a^*$

$S_2 \to S_2 b / \epsilon \Rightarrow b^*$

} LLG

Language ⟶ ? grammar

Grammar ⟶ ? language

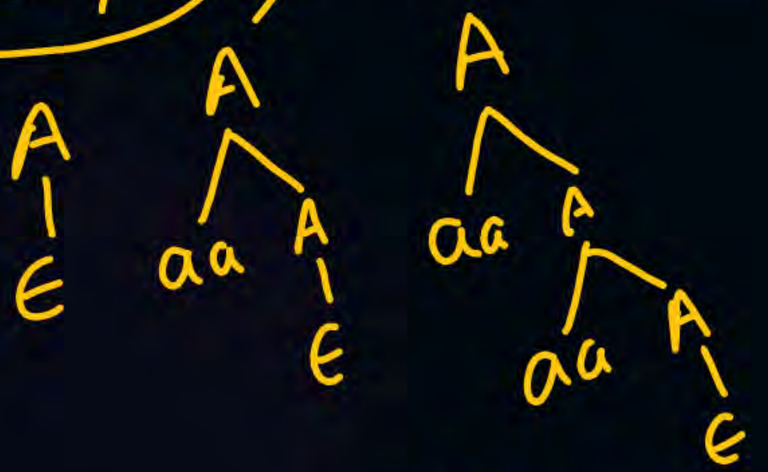
CFG

$S \to A/B$

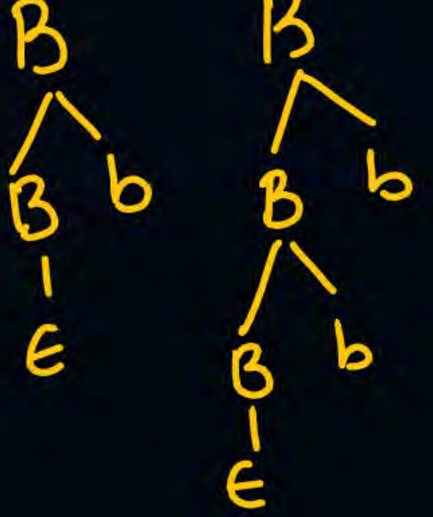
$A \to aaA/\epsilon$

$B \to Bb/\epsilon$

not Regular



$(aa)^*$

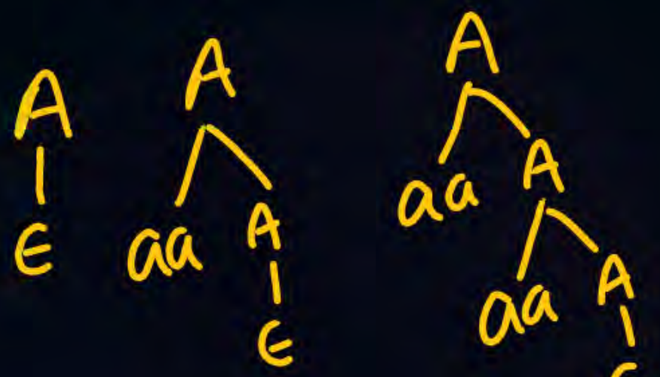


$(b)^*$

$(aa)^* + b^*$ → Regular

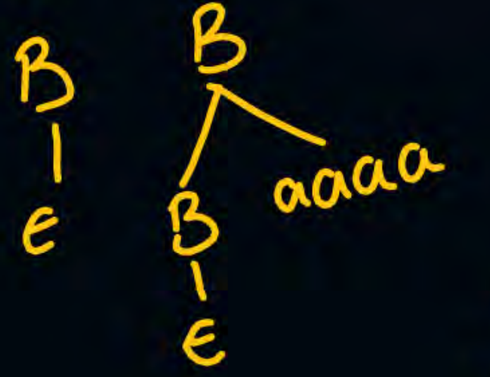
$S \to A/B$
$A \to aaA/\epsilon$
$B \to Baaaaa/\epsilon$

} not regular grammar



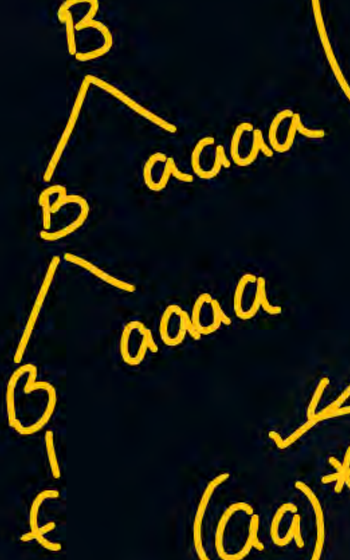
$A \Rightarrow (aa)^*$

a's divisible by 2

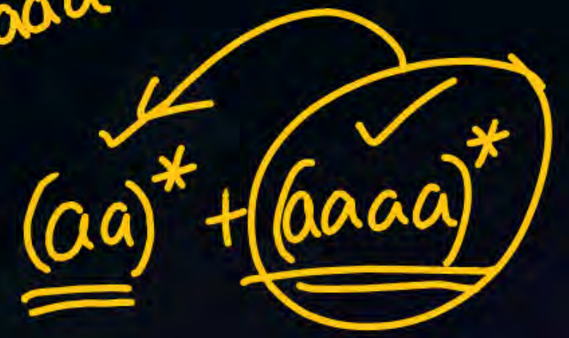


$B \Rightarrow (aaaa)^*$

a's divisible by 4







$(aa)^*$

a^*b^* $\Rightarrow$ FA $\Rightarrow$ RLG

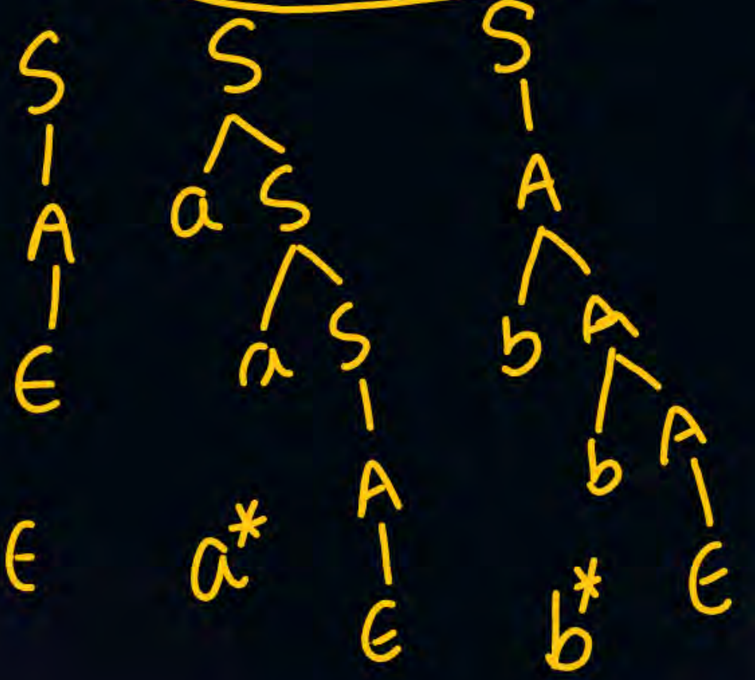
ϵ, a^*, b^*, a^*b^* ✓

CFG { $S \to AB$, $A \to aA/\epsilon$, $B \to bB/\epsilon$ } Not Reg grammar

$A \to \alpha B/\beta$

$\alpha, \beta \to$ Terminals Σ^*

$S \to aS/A$
$A \to bA/\epsilon$





$S \to aS/A$
$A \to bA/\epsilon$

$A \Rightarrow b^*$

$S \Rightarrow a^* S$
$S \Rightarrow a^* A$
$S \Rightarrow a^* b^*$

$a^* b^*$

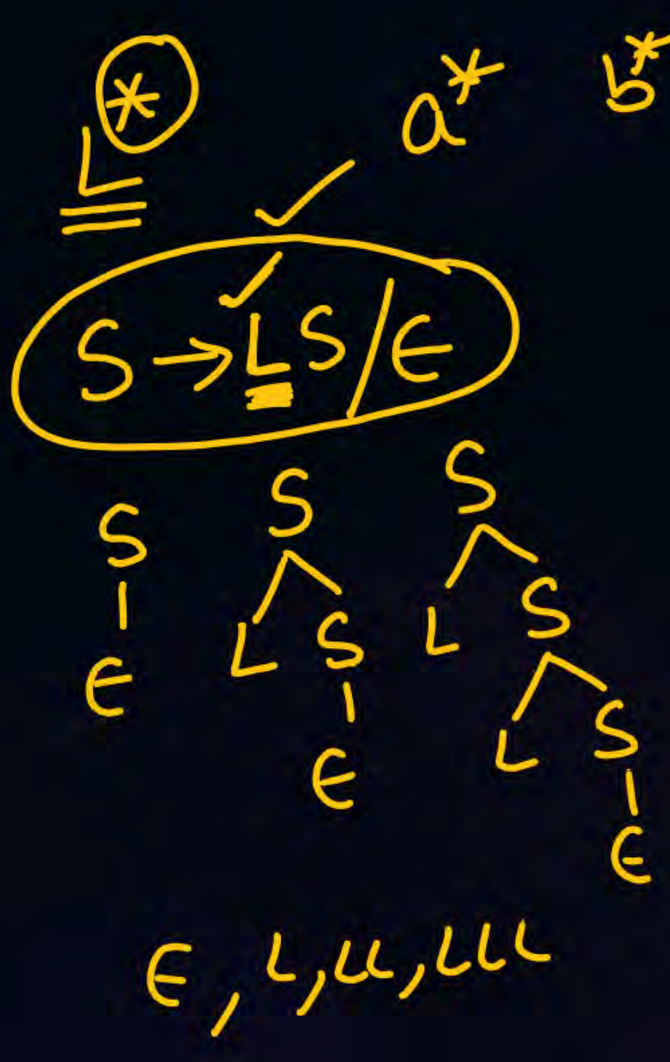
$S \to aS/bA/\epsilon$

$(a+b)^*$

$S \to a \mid b \mid SS \mid \epsilon$

$S \to aS \mid bS \mid \epsilon$

$S \to Sa \mid Sb \mid \epsilon$



$S \rightarrow (a+b)S / \epsilon$

$S \rightarrow aS / bS / \epsilon$

$(\underbrace{00}_{a} + \underbrace{110}_{b})^*$

$S \to 00/110/SS/\epsilon \to$ CFG

$S \to 00S/110S/\epsilon$

$S \to S00/S110/\epsilon$ } Reg

$(a+b)^+$

$S \to \check{a} / \check{b} / SS \to$ CFG

$S \to aS / bS / a / b$

RLG

$S \to Sa / Sb / a / b$

LLG

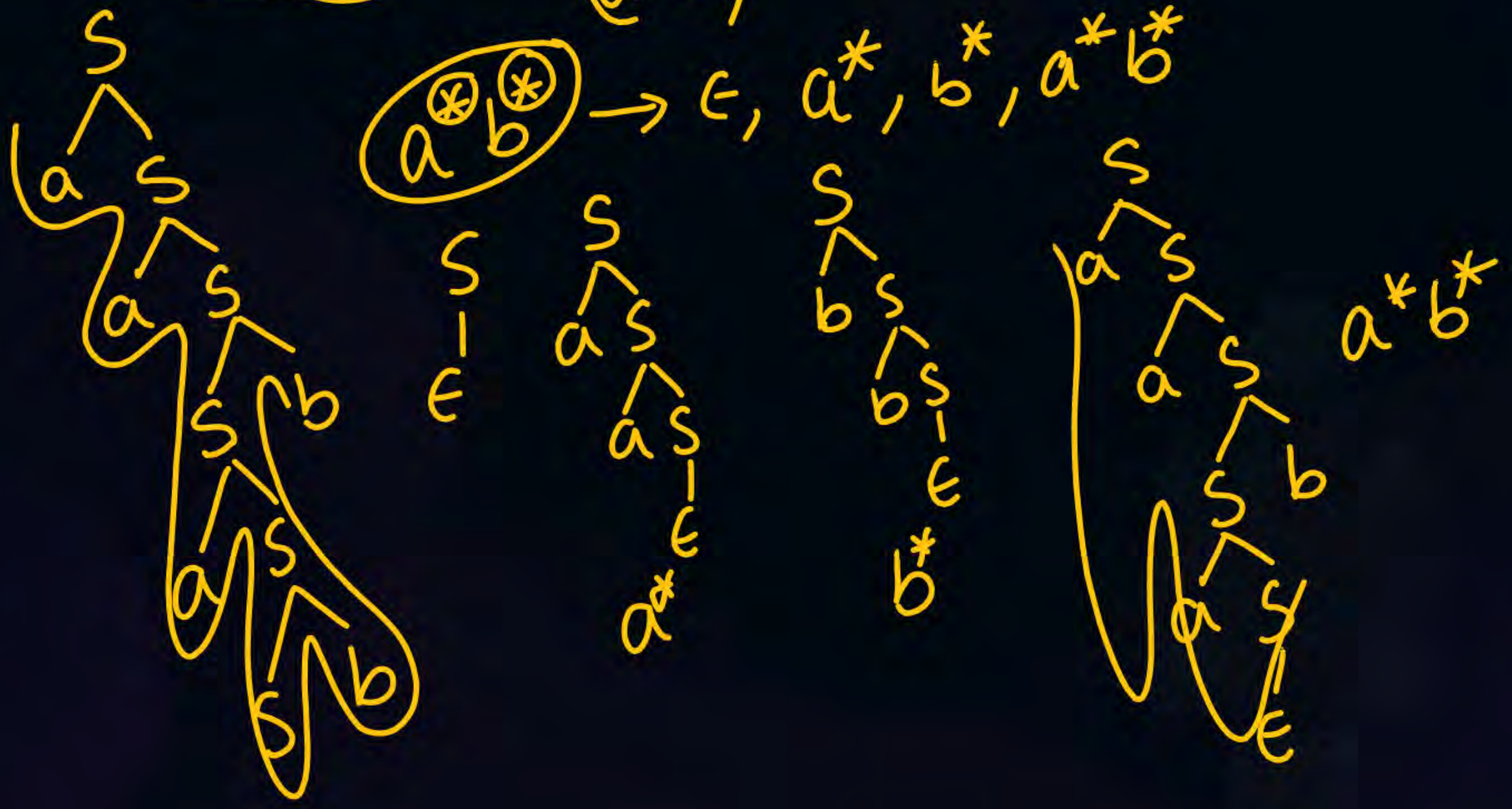
} Reg grammars

$S \to aS \mid bS \mid \epsilon \Rightarrow (a+b)^*$

Sb

(a^*b^*)

$a^*b^* \to \epsilon, a^*, b^*, a^*b^*$



$S \to aS/bS/\epsilon \Rightarrow (a+b)^*$

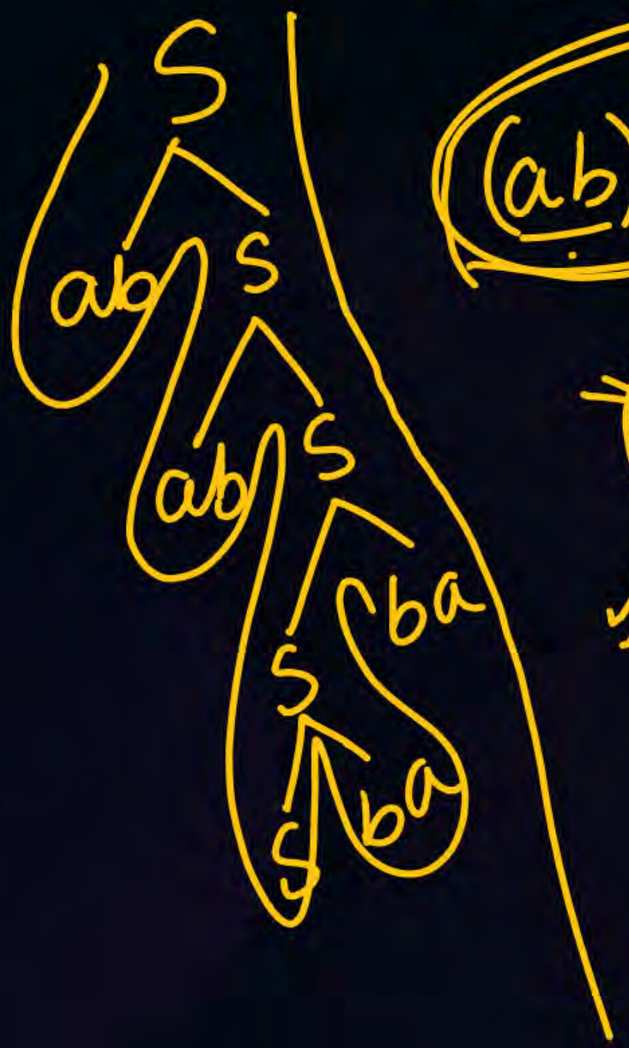
Sa



b^*a^*

$S \to ab\,S \mid ba\,S \mid \epsilon \Rightarrow (ab+ba)^*$

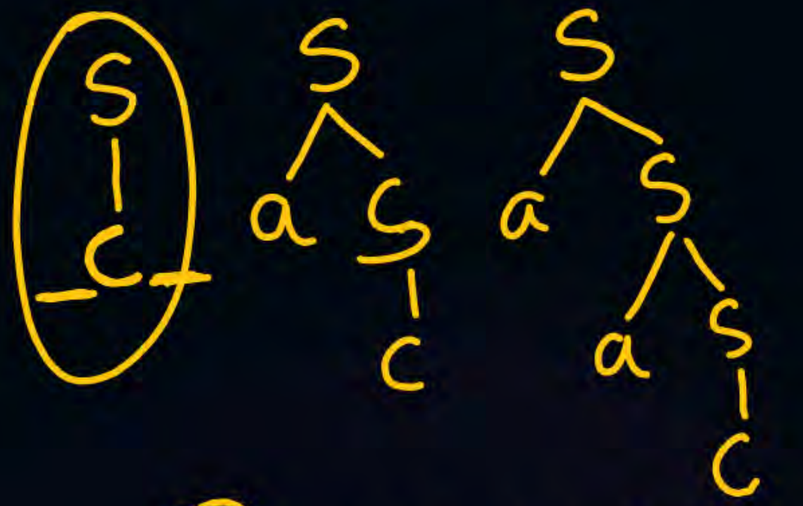
Sba



$(ab)^*(ba)^*$

$a^*b^* \to \epsilon, a^*, b^*, a^*b^*$

$S \to aS/Sb/c$

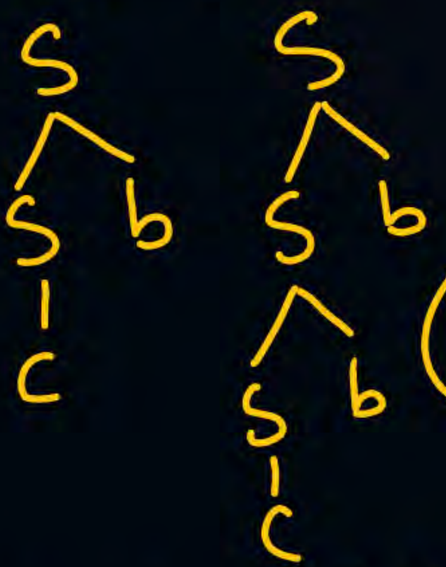


c + ac + aac ...

a^*c ✓

a, b, c → Terminal

S → Variable

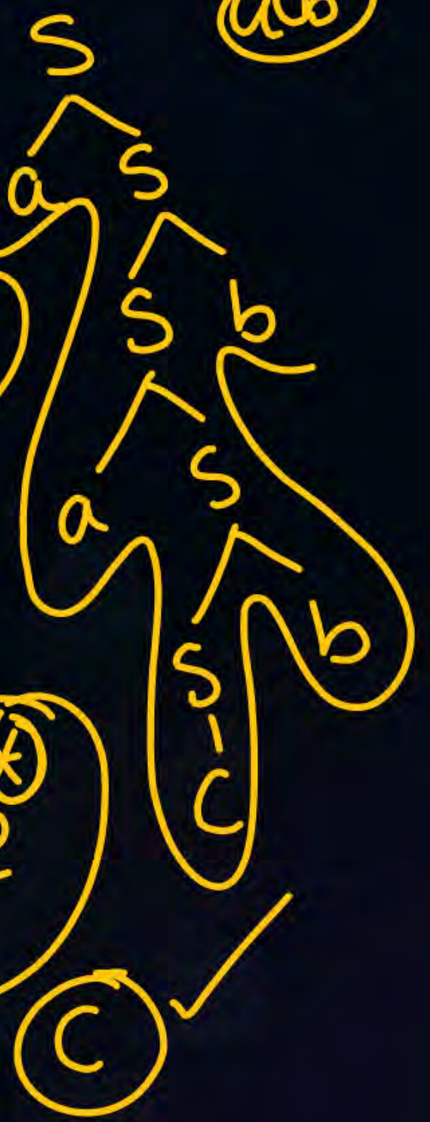


c + cb + cbb + cbbb

cb^* ✓

a^+cb^+

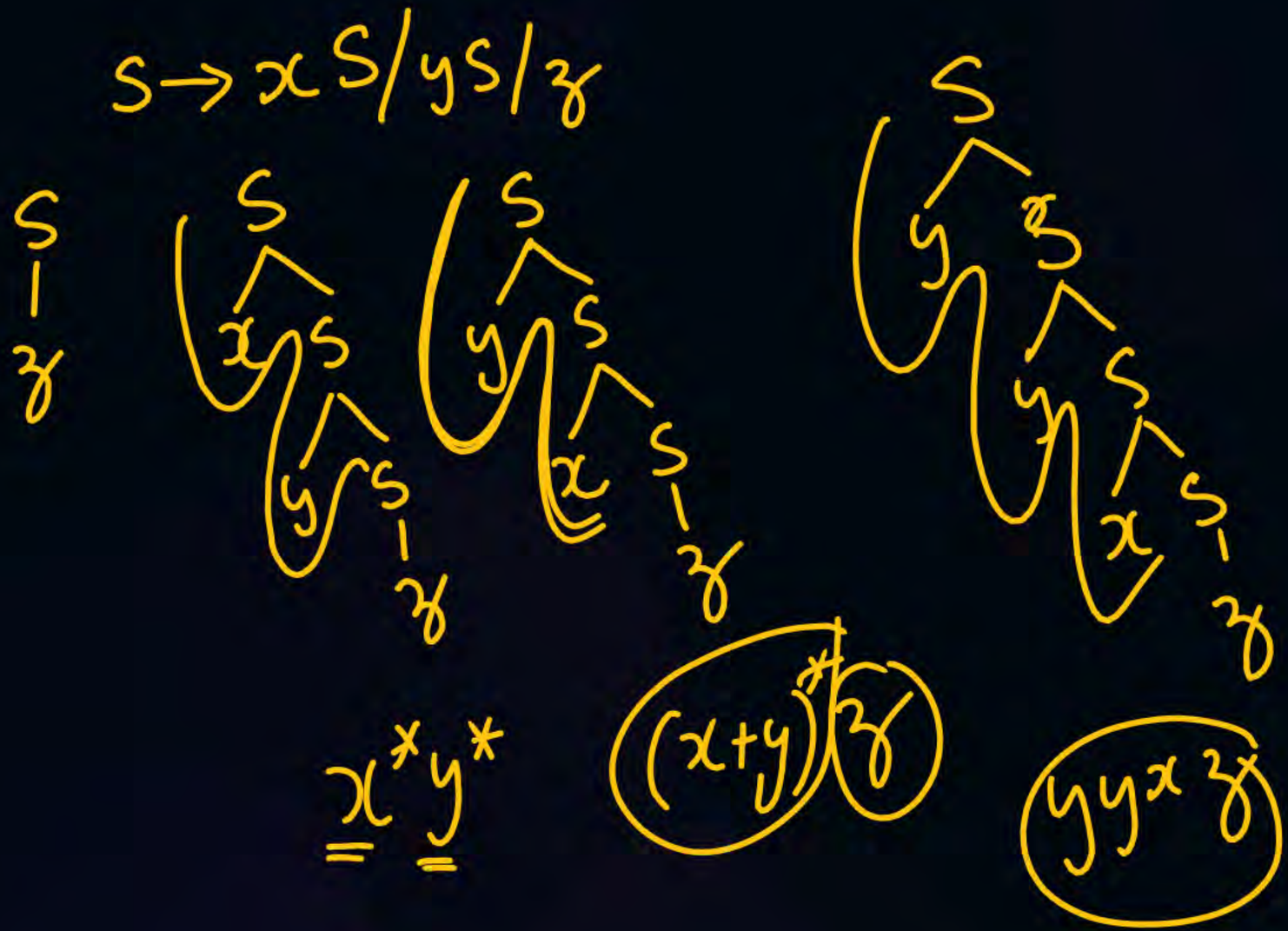
acb



a^*cb^*

a^*cb^* ✓

$S \rightarrow xS / yS / z$



$x^* y^*$

$(x+y)^* z$

$yyxz$

$S \to Sx / Sy / z$



zyx

zxy

$ab^* \neq b^*a$

$(a+b)^*$ a^* b^*

$(a+b)^*$

$z(x+y)^*$

$z(x^*y^*)^*$

$aa^* = a^*a = a^+$

$(a+b)^* a a^*$

$(a+b)^* a^* a$

$(a+b)^* a$

$(a+b)^* a^* = (a+b)^*$

$(a+b)^* b a^*$

atleast one 'b'

$ba^* \neq a^*b$

$(a+b)^* b (a+b)^*$

$(a+b)^* b a^*$

$a^* b (a+b)^*$

$a^* b a^*$

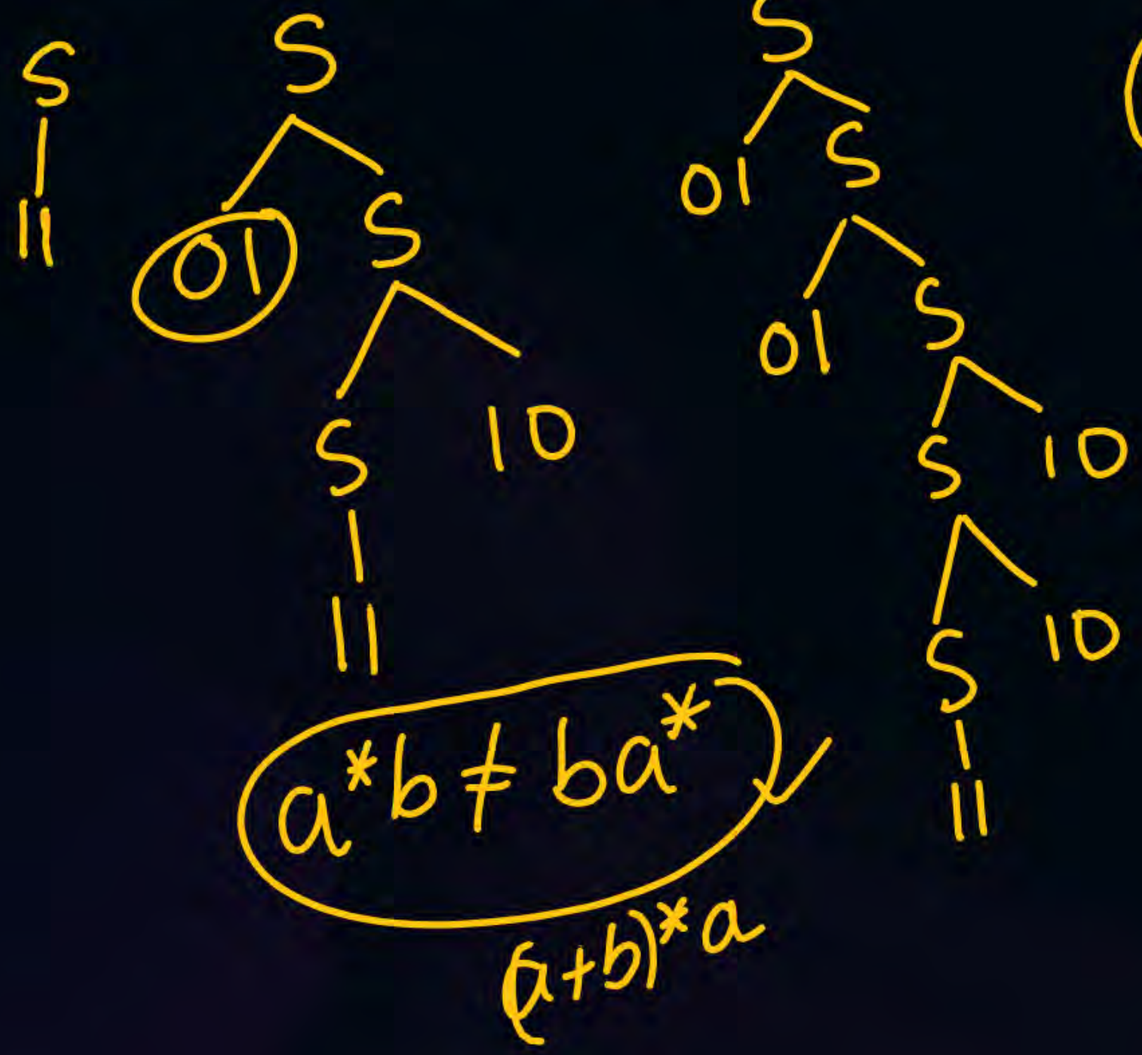
exactly 1b

$(a+b)^* b a^*$

$\neq (a+b)^* b$

$(a+b)^* b$

$S \rightarrow 01S / S10 / 11$



$(01)^* 11 (10)^*$

$L = \{a, b\}$

$A \to a/b$

L = Strings of length '2'

$S \to aa/ab/ba/bb$

$(a+b)(a+b)$ ✓

$S \to AA$
$A \to a/b$

NFA



$A \to aB/bB$
$B \to aC/bC$
$C \to \in$

RLG

$A \to aB/bB$
$B \to a/b$

RLG

String length of exactly '2'



$A \to aB/bB$

$B \to aC/bC$

$C \to \epsilon$

RLG

$A \to aB/bB$

$B \to a/b$

at least 2

$\underbrace{(a+b)}_{B}\underbrace{(a+b)}_{B}\underbrace{(a+b)^*}_{A}$

CFG
$S \to BBA$
$B \to a/b$
$A \to aA/bA/\epsilon$

Non regular

RLG
$A \to aB/bB$
$B \to aC/bC$
$C \to aC/bC/\epsilon$



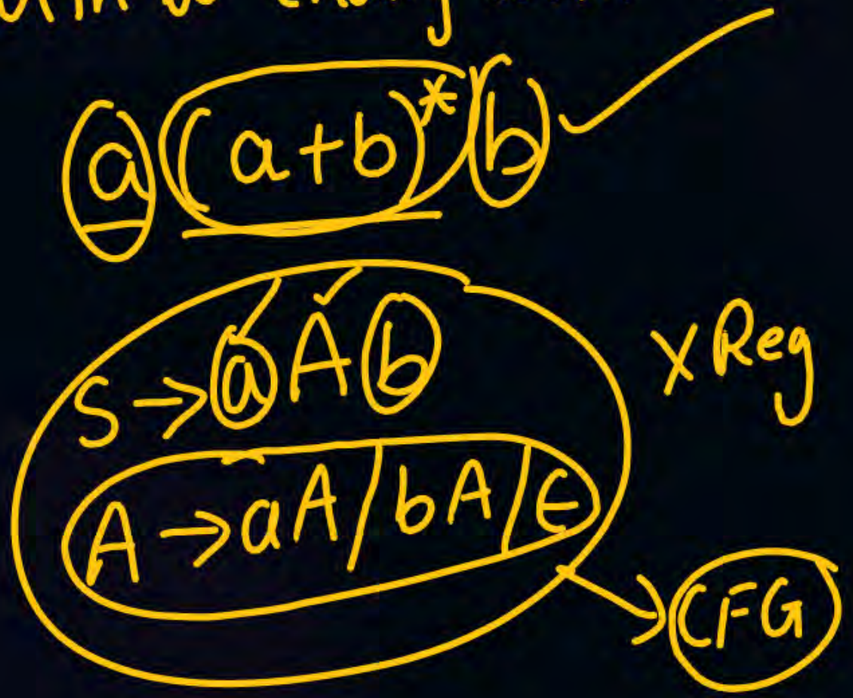
String length of atmost 2

$(a+b+\epsilon)(a+b+\epsilon)$

$S \rightarrow AA$

$A \rightarrow a/b/\epsilon$

Starting with 'a' ending with 'b'



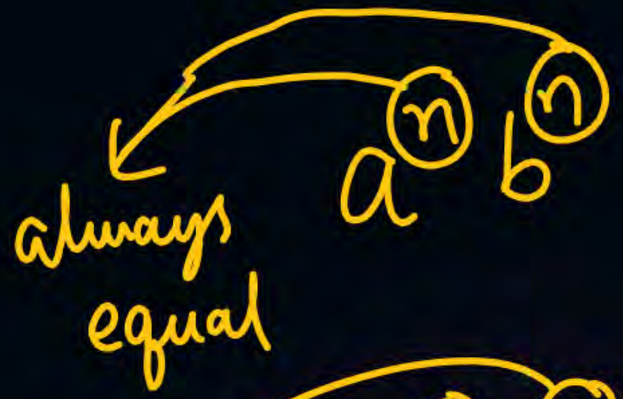
Starting and ending with diff symbols

$$a(a+b)^* b + b(a+b)^* a$$

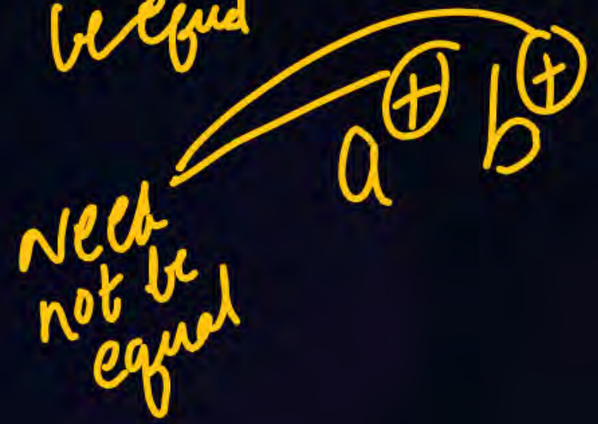
$S \rightarrow aAb / bAa$

$A \rightarrow aA / bA / \epsilon$

$a^n b^n / n \geq 1$







$a^* b^*$

$a^+ b^+$

$a^2 b^3$

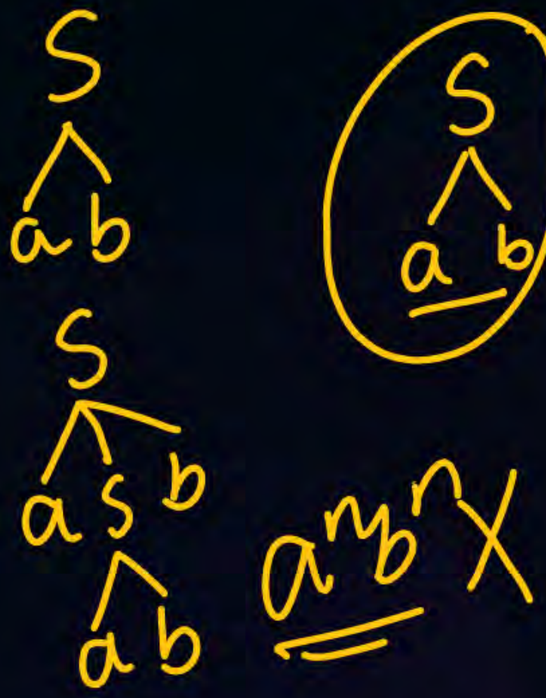
$a^3 b^2$

$a^n b^n \subseteq a^* b^*$

$a^n b^n / n \geq 1$ → This language is not regular

CFL

$S \rightarrow aSb / ab$ → CFG



$a^n b^n$ ✗

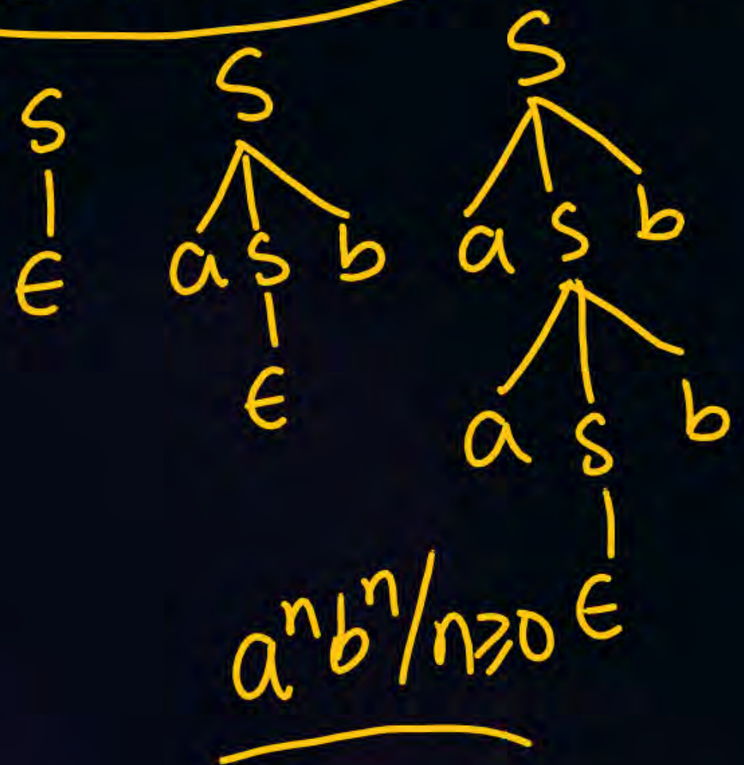
FA ✗

RE ✗

Reg gramm ✗

$a^n b^n / n \geq 0$ → lang not Regular → CFL

$S \to aSb / \epsilon$ → grammar not Regular



$a^n b^n / n \geq 0$

$a(ab)^* b$

$a(ababab)b$

$a^n b^m / n, m \geq 1$

$a^+ b^+$

$S \rightarrow AB$
$A \rightarrow aA / a$
$B \rightarrow bB / b$

$a^n b^n c^m / n, m \geq 1$

$a^n c^m b^n / n, m \geq 1$

$a^n b^n c^m d^m / \cancel{A, AB,} + m, n \geq 1$

$a^n b^n c^n / n \geq 1$

$a^n b^{2n} / n \geq 1$

THANK - YOU